

NORMANTON, Australia

WMO: 942660

Lat: 17.6869S

Lon: 141.0733E

Elev: 19

StdP: 101.10

Time Zone: 10.00 (E10)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	14.9	-7.1	2.1	24.2	-5.0	2.5	23.4	9.4	19.3	8.8	20.0	4.4	170	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	9.8	37.5	21.6	36.5	22.5	35.7	23.0	28.1	31.3	27.5	31.0	27.1	30.7	4.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.3	23.2	29.3	26.7	22.3	29.1	26.2	21.7	28.9	90.2	31.5	87.7	30.9	85.8	30.8	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.6	7.0	DB	11.6	40.2	1.3	1.2	10.6	41.0	9.8	41.7	9.1	42.4	8.1	43.2
			WB	5.6	28.8	0.8	1.2	5.0	29.7	4.5	30.4	4.1	31.1	3.5	32.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	29.3	29.2	29.1	28.2	25.5	23.1	22.9	24.1	27.2	29.6	30.5	30.5
	DBStd	3.35	1.84	1.62	1.61	1.79	2.32	2.70	2.46	2.17	1.78	1.49	1.49	1.74
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	4	0	0	0	0	0	2	2	0	0	0	0	0
	CDD10.0	6351	597	538	591	546	479	392	398	436	516	609	615	634
	CDD18.3	3313	339	305	332	296	221	144	142	178	266	351	365	376
	CDH23.3	38424	4151	3717	4028	3413	2076	1170	1181	1644	2832	4389	4829	4994
(13)	CDH26.7	18246	1921	1690	1840	1586	831	379	372	648	1403	2323	2594	2659
(14)	Wind	WSAvg	3.4	3.1	2.8	2.8	3.0	3.4	3.8	3.7	3.8	3.8	3.7	3.5
(15) Precipitation	PrecAvg	853	229	214	156	27	6	7	2	1	5	12	46	138
	PrecMax	1852	965	510	475	121	39	85	22	14	63	79	232	402
	PrecMin	444	42	12	4	0	0	0	0	0	0	0	0	31
	PrecStd	341	190	130	100	32	11	19	4	3	12	19	47	93
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	36.7	36.6	35.9	34.3	32.4	32.3	34.4	36.5	38.4	38.9	39.0
		MCWB	22.9	25.1	23.0	22.3	21.4	20.3	20.5	20.2	19.4	20.0	19.1	21.6
	2%	DB	35.7	35.3	35.1	34.7	32.9	31.2	31.0	32.7	35.2	36.8	37.2	37.1
		MCWB	25.0	25.3	24.2	22.4	21.0	19.9	19.4	18.9	19.9	20.9	21.2	23.1
	5%	DB	34.5	34.3	34.2	33.9	31.9	30.2	30.0	31.4	34.1	35.7	36.1	35.9
		MCWB	25.7	25.5	24.3	22.1	20.6	19.5	18.5	18.3	19.8	21.3	22.5	24.2
	10%	DB	33.4	33.3	33.2	33.0	30.9	29.1	29.0	30.2	33.0	34.8	35.1	34.9
		MCWB	25.8	25.7	24.3	22.0	20.2	18.8	17.9	17.7	19.5	21.4	23.3	24.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.4	29.4	27.8	26.5	24.4	22.5	22.2	22.5	24.4	27.1	27.8	28.1
		MCDB	31.1	32.3	30.7	30.5	29.9	28.8	29.6	30.3	31.4	32.7	31.5	31.5
	2%	WB	28.2	27.8	27.2	25.7	23.4	21.5	21.1	21.4	23.5	25.9	27.0	27.5
		MCDB	31.1	31.2	30.4	29.4	29.1	28.4	28.5	28.6	30.1	31.2	31.3	31.0
	5%	WB	27.5	27.3	26.7	25.2	22.6	20.7	20.2	20.3	22.9	25.2	26.4	27.0
		MCDB	30.9	30.7	30.0	29.1	28.6	27.5	27.3	27.9	29.1	30.2	30.8	30.7
	10%	WB	27.0	26.9	26.2	24.6	21.7	19.9	19.2	19.4	22.2	24.6	25.9	26.6
		MCDB	30.7	30.4	29.7	28.9	28.2	27.2	26.7	27.2	28.4	29.8	30.5	30.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.3	7.9	9.6	10.8	11.4	11.9	12.8	12.5	11.3	9.8	8.7
		MCDBR	9.1	9.2	9.5	10.5	11.3	11.7	12.2	13.4	13.3	12.3	11.1	10.3
		MCWBR	3.0	2.9	3.1	3.6	4.4	5.3	5.5	5.4	4.8	4.5	4.2	3.6
	5% WB	MCDBR	7.6	7.6	8.0	8.1	9.5	10.8	11.2	12.4	11.4	10.0	9.0	8.2
		MCWBR	3.0	2.8	2.9	3.0	4.0	5.0	5.0	5.0	4.5	3.5	3.4	3.0
(40) Clear-Sky Solar Irradiance	taub		0.413	0.409	0.387	0.363	0.344	0.330	0.329	0.344	0.391	0.419	0.444	0.438
	taud		2.482	2.495	2.550	2.575	2.556	2.557	2.539	2.474	2.356	2.310	2.297	2.358
	Ebn at Noon		932	926	926	912	894	890	901	916	906	906	898	909
	Edh at Noon		118	115	105	96	91	88	91	104	125	137	141	133
(44) All-Sky Solar Radiation	RadAvg		5.89	6.06	6.06	5.83	5.05	4.64	4.97	5.78	6.68	7.17	7.04	6.53
	RadStd		0.61	0.60	0.55	0.41	0.25	0.31	0.17	0.19	0.24	0.37	0.41	0.58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page